

Project Title

Preventing Wrong Decisions in Smart Building Systems

Mentor Name

Project Number

S.No	Rollnumber	Branch	Name	Role	Signature
1.	R2142232076	B.Tech CSE/CSF	Rishi Kumar	Architecture	
2.	R2142230443	B.Tech CSE/CSF	Pranav Venkatraman	IOT Integration	
3.	R2142231346	B.Tech CSE/CSF	Aman Rajpoot	Attach Simulation and Defense	
4.	R2142231273	B.Tech CSE/CSF	Amalkrishnan Ajay	Python Encoder & ML	

Project Mentor

Cluster Head

Date	Understanding of Project	Project Working	Soft Skills	Report	Mentor Marks	Total Marks	Project Status
R.No	25 Marks	35 Marks	10 Marks	15 MARKS	85 MARKS	100 MARKS	Activity Coordinator

Synopsis Evaluation

Theoretical Understanding

Rollno	Problem(4 Marks)	Algorithm(4 Marks)	Data /Data structure(4 Marks)	SWOT Analysis(4 Marks)	Area of Application(4 Marks)	Total Marks(20)

Panel Remark

Reviewer 1

Reviewer 2

Reviewer 3

Reviewer 4

Reviewer 5

Mid- Term Evaluation

DESIGN & DEVELOPMENT

Rollno	Technical Diagram(5 Marks)	Programming Concepts(5 Marks)	IPC(5 Marks)	Libraries(5 Marks)	SRS(10)	Total(20 Marks)

Panel Remark

Reviewer 1

Reviewer 2

Reviewer 3

Reviewer 4

Reviewer 5

End-Term Evaluation

Testing & Implementation

Rollno	Theoretical Knowledge(5)	Computational Knowledge(5)	Test Case (10)	Soft Skills (10)	Report(5)	Core Computational Skills(15)	Total (50)

Panel Remark

Reviewer 1

Reviewer 2

Reviewer 3

Reviewer 4

Reviewer 5

Project Title							Mentor Name			
Abstract	Smart building systems rely on sensors to automate decisions but often trust incorrect data without verification. Small sensor errors can silently cause energy waste and discomfort without triggering alarms. This project detects such issues by comparing multiple sensor readings and identifying inconsistent behavior. The approach is low-cost, explainable, and compatible with existing systems.									
Objective	The goal is to prevent wrong automation decisions caused by faulty or manipulated sensor data. The system verifies sensor reliability using majority agreement instead of single readings. It detects silent sensor drift and believable errors. This ensures safer and more efficient building operation.									
Methodology	Multiple ESP32 sensors monitor the same environment and send data through MQTT. A central system compares readings to form a consensus baseline. Sensors that deviate over time lose trust and are flagged. Decisions are accepted only when most trusted sensors agree.									
Progress 1										
Mentor Remark	Marks	10	10	10	10	10	10	10	15	
	Roll Number	Step 1	Step 2	Step 3	Step 4	Step 5	Step 6	Step 7	Internal	
	Date/Mentor Signature									
Progress 2										
Mentor Remark	Marks	10	10	10	10	10	10	10	15	
	Roll Number	Step 1	Step 2	Step 3	Step 4	Step 5	Step 6	Step 7	Internal	
	Date/Mentor Signature									

Guideline: 1) A project group can be of maximum 4 members and no alteration in the group member will be entertained later.

Guideline: 2) Methodology should have following steps Step1: Literature Review; Step2: Identification Of Requirement (Type Of Data source, Amount Of Data, & Format of Data); Step3: Identification of Algorithm; Step4 : Comparative study; Step5: Design and Development of System/Architecture; Step 6: Implementation; Step7: Results

Guideline:3) Student should upload softcopies of all the documents (reports and power point presentations) in "Project Directory", 24 hrs prior to evaluation.

Guideline:4) Panel member will give feedback to individual on the scale of 1 to 5 and this scale will change for defaulter i.e. 1 to 3 scale.

1: Poor 2: Average 3: Good 4: Excellent 5: Outstanding